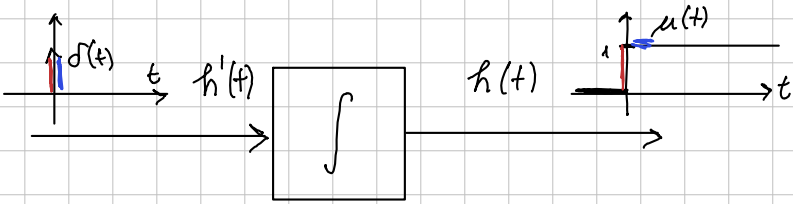
KONVOLUCIJSKI INTEGRAL

Odziv kont. lin. sustava

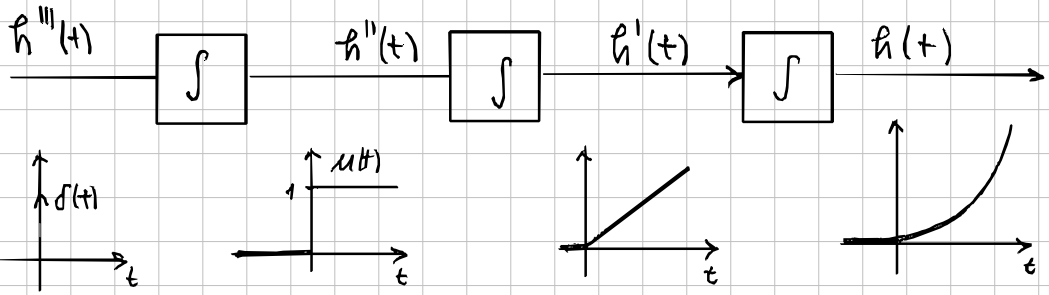
Kako dođe do impulznog odziva? * pomoćna metoda

integrator: $y(t) = \int_{-\infty}^{t} u(\tau)\,d\tau = y(0^-) + \int_{0^-}^{t} u(\tau)\,d\tau$



$h'(0^-) = 0$ $\qquad$ $h(0^-) = 0$

$h'(0^+) = 0$ $\qquad$ $h(0^+) = 1$



$h(0^-) = 0$ $\qquad$ $h(0^-) = 0$ $\qquad$ $h(0^-) = 0$ $\qquad$ $h(0^-) = 0$

$h(0^+) = 0$ $\qquad$ $h(0^+) = 1$ $\qquad$ $h(0^+) = \cancel{1}\ 0$ $\qquad$ $h(0^+) = \cancel{1}^2\ 0$

inače nema promjene poč uvjeta

jer uvrštavamo $t=0$ u tu funkciju!

Sustav koji promatramo u kontinuiranom vremenu obilježavamo diferencijalnim jednadžbama.

$$y'(t)+2y(t) = u(t)$$

$$u(t)=\delta(t)$$

Sustav koji promatramo u kontinuiranom vremenu obilježavamo diferencijalnim jednadžbama.

potrebni poč. uvjeti: $y(0^-)=0$

1° $t<0$ sustav miran, nema impulsa, ni ulaz i poč vr. $=\emptyset$

$\longrightarrow y(t)=0 \Longrightarrow h(t)=0$ (jer tražimo impulsni odziv)

2° $t>0$

$h'(t)+2h(t)=0$ ← jer $u(t)=\delta(t)$ koji je bočno u $\emptyset$ jednak 1

$h'(t)=-2h(t) \longrightarrow h(t)=C\cdot e^{-2t}$

3° $t=0$ $\quad h'(t)+2h(t)=\delta(t) \quad \Big/ \int_{0^-}^{0^+}$

$$\int_{0^-}^{0^+} h'(t)\,dt + 2\int_{0^-}^{0^+} h(t)\,dt = \underbrace{\int_{0^-}^{0^+}\delta(t)\,dt}_{1}$$

$$h(t)\Big|_{0^-}^{0^+} + 2\cdot 0 = 1$$

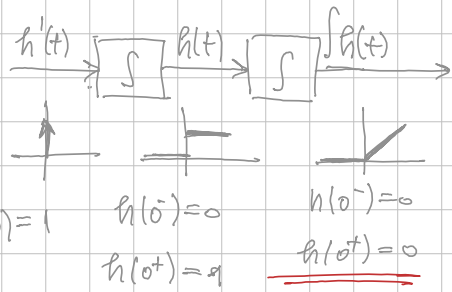
$h(0^+)-\cancel{h(0^-)}$

sustav je miran $\rightarrow h(0^-)=\emptyset$

$\Rightarrow$ $h(0^+)=1$

$h(0^+)=C\cdot e^{-2\cdot 0}=1 \rightarrow C=1$

$$\boxed{h(t)=1\cdot e^{-2t}u(t) \text{ jer kreće tek od } \emptyset!}$$



Kako sad do konvolucije?

$$u(t)*\delta(t)=\int_{-\infty}^{\infty}u(\tau)\delta(t-\tau)\,d\tau = u(t)$$

$\tau=t$

ako sad taj ulazni signal stavimo u sustav, dobit ćemo $y(t)$

$y(t)=S(u(t))$

↳ ako tražimo impulsni odziv: $h(t)=S(\delta(t))$

Nas zanima koliko iznosi odziv na bilokoji ulaz ako znamo da je impulsni odziv h.

$$y(t)=S(u(t))=S\left(\int_{-\infty}^{\infty}u(\tau)\delta(t-\tau)\,d\tau\right)$$

$$=\int_{-\infty}^{\infty}u(\tau)\underbrace{S(\delta(t-\tau))}_{h(t-\tau)}\,d\tau$$

(pomaknuti impulsni odziv)

$$=\int_{-\infty}^{\infty}u(\tau)h(t-\tau)\,d\tau \quad = \underline{\text{def. konvolucije}}$$

$$\boxed{y(t)=u(t)*h(t)}$$

SVOJSTVA : Konvolucije

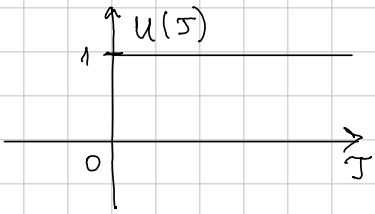
1. Komutativnost $\quad x_1 * x_2 = x_2 * x_1$
2. Distributivnost $\quad x_1 * (x_2 + x_3) = x_1 * x_2 + x_1 * x_3$
3. Asocijativnost $\quad x_1 * (x_2 * x_3) = (x_1 * x_2) * x_3$
4. Pomak $\quad x_1(t-\tau_1) * x_2(t-\tau_2) = y(t-\tau_1-\tau_2)$
5. S impulsom $\quad x(t) * \delta(t) = x(t)$

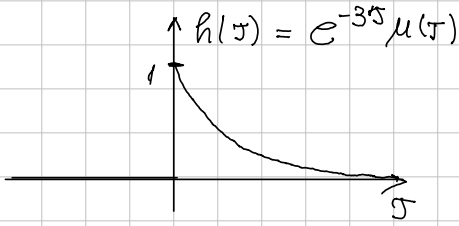
Primjer: konvolucija dva signala

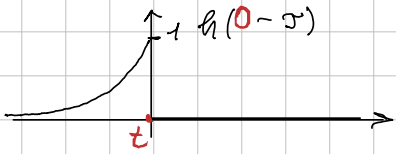
$h(t) = e^{-3t}\mu(t)$

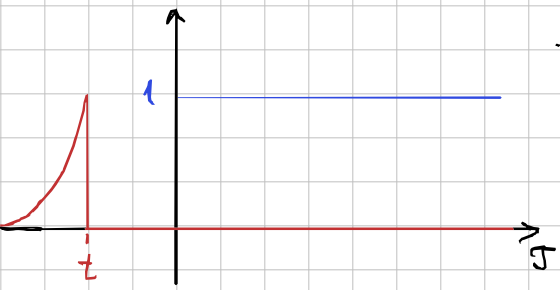
$u(t) = \mu(t) \qquad y(t) = ?$

$$y(t) = \int_{-\infty}^{\infty} u(\tau)h(t-\tau)d\tau = \int_{-\infty}^{\infty} u(t-\tau)h(\tau)d\tau$$



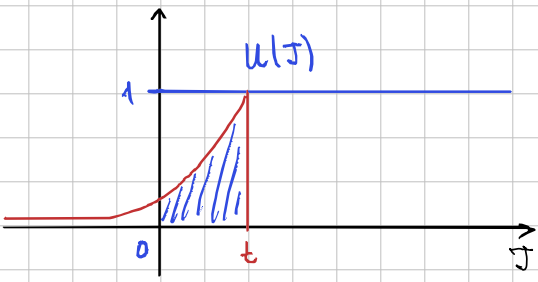






tamo di je jedan ≠0 drugi je =0

$\rightarrow y(t) = 0, \quad t<0$



$t>0$

$$y(t) = \int_0^t 1 \cdot e^{-3(t-\tau)} d\tau$$

$$= e^{-3t}\int_0^t e^{3\tau} d\tau = e^{-3t} \left.\frac{e^{3\tau}}{3}\right|_0^t$$

$$y(t) = \frac{e^{-3t}}{3}\left(e^{3t} - e^0\right) = \frac{1}{3} - \frac{e^{-3t}}{3}$$

u ovom zadatku imamo samo ove dvije situacije

RJ:

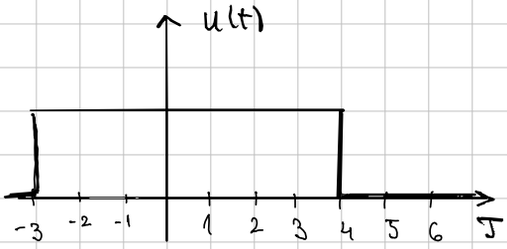
$$\boxed{\Rightarrow y(t) = \left[\frac{1}{3} - \frac{e^{-3t}}{3}\right]\mu(t)}$$

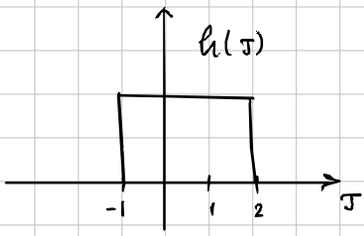
Primjer: $h(t) = \mu(t+1) - \mu(t-2)$

$u(t) = \mu(t+3) - \mu(t-4)$

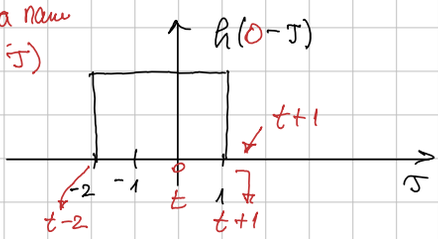
$y(t) = ?$

FORMULA: $y(t) = \int_{-\infty}^{\infty} u(\tau)\, h(t-\tau)\, d\tau$ (i da obrnemo, rez mora biti isti)





treba nam $h(-\tau)$



1°

$y(t) = 0$

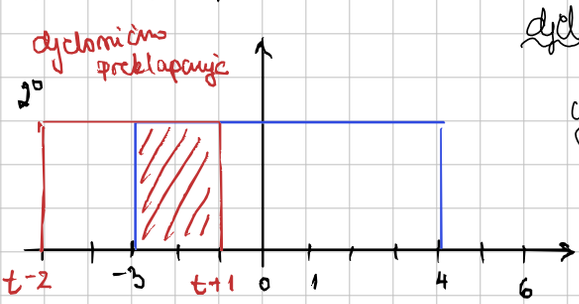
Presjeka nema sve dok $t+1 < -3$

završetak crvenog ne talasa $t < -4$

početak plavog

dijelomično preklapanje

2°



dok $(t-2)$ ne uđe pod plavi → potpuno preklapanje

dijelomično preklapanje

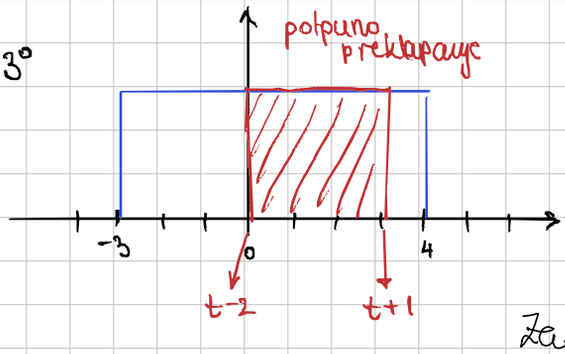
$$y(t) = \int_{-3}^{t+1} 1 \cdot 1\, d\tau = \tau \Big|_{-3}^{t+1} = t+1+3$$

$y(t) = t+4$ *(za koji dio vremena?)

Za $t \geq -4$

kraj: $t-2 = 3 \longrightarrow t = -1$

3°



potpuno preklapanje

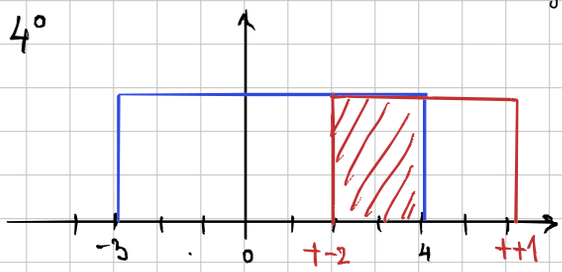
$$y(t) = \int_{t-2}^{t+1} 1 \cdot 1\, d\tau = \tau \Big|_{t-2}^{t+1}$$

$y(t) = \cancel{t}+1-\cancel{t}+2 \longrightarrow y(t) = 3$

Za $t-2 = -3 \longrightarrow t \geq -1$

kraj $t+1 = 4 \longrightarrow t \leq 3$

4°



Djelomično preklapanje

$$y(t) = \int_{t-2}^{4} 1 \cdot 1\, d\tau = \tau \Big|_{t-2}^{4}$$

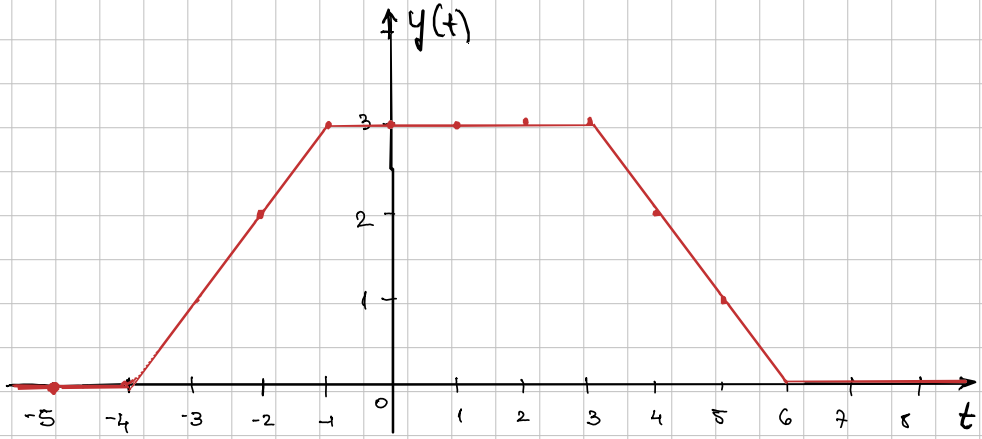
$y(t) = 4 - t + 2 = 6 - t$

Za $t+1 = 4 \longrightarrow t = 3$

kraj $t-2 = 4 \longrightarrow t = 6$

5° kada crveni ~~skroz~~ izađe

$y(t) = 0$

konačno rješenje:

$$y(t) = \begin{cases} 0, & t < -4 \\ t+4, & t \in [-4, -1\rangle \\ 3, & t \in [-1, 3\rangle \\ 6-t, & t \in [3, 6\rangle \\ 0, & t \geq 6 \end{cases}$$